Lecture Plan: Final Examination, Spring 2019-20

PHYSICS 2 [1203]: Credit: 3 [24 lectures (36 hours)]
Heat and Thermodynamics & Oscillations, Waves and Optics
Textbook: Fundamentals of Physics (10th edition, Extended): David Halliday, Robert Resnick and Jearl Walker (John Wiley & Sons)

Final Examination [60%]: Oscillations, Waves and Optics [12 lectures (18 hours)]

1st Lecture:
Description of the course

Objective:
1. In simple harmonic motion (SHM), the displacement *x(t)* of a particle from its equilibrium position is described by the equation $x = x_m \cos(\omega t + \varphi)$. (i) Derive equations for the particle's SHM velocity and acceleration as functions of time. (ii) Hence find the maximum displacement amplitude, the maximum velocity amplitude and the maximum acceleration amplitude. (iii) Sketch a graph of an oscillator's displacement versus time, velocity versus time and acceleration versus time.
2. In simple harmonic motion, the displacement *x(t)* of a particle from its equilibrium position is described by the equation $x = x_m \cos(\omega t + \varphi)$. From this, prove that the time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{m}{k}}$.

Chapter 15: Oscillations
15-1: Simple harmonic motion (simple harmonic motion, the velocity of SHM, the acceleration of SHM, the force law for simple harmonic motion)

Application:
Module 15-1 Simple Harmonic Motion
3. What is the maximum acceleration of a platform that oscillates at amplitude 2.20 cm and frequency 6.60 Hz?
13. An oscillator consists of a block of mass 0.500 kg connected to a spring. When set into oscillation with amplitude 35.0 cm, the oscillator repeats its motion every 0.500 s. Find the (a) period, (b) frequency, (c) angular frequency, (d) spring constant, (e) maximum speed, and (f) magnitude of the maximum force on the block from the spring.

2nd Lecture:
Objective:
3. The mechanical energy of a linear oscillator is indeed constant and independent of time, $E = K + U = \frac{1}{2} k x_m^2$. Sketch a graph of the kinetic energy, potential energy, and total energy of a spring–block oscillator, first as a function of time and then as a function of the oscillator's position.

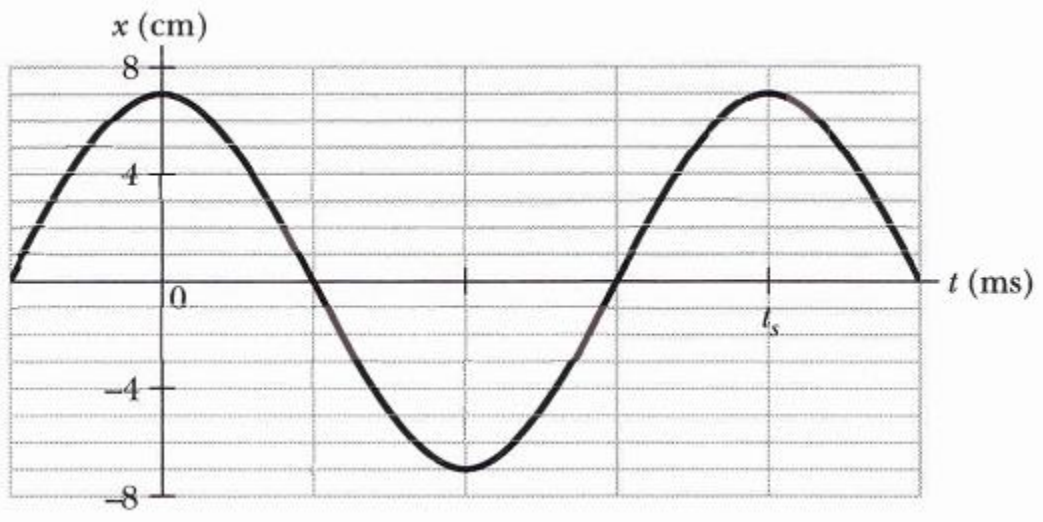
Chapter 15:
15-2: Energy in simple harmonic motion

Application:
Module 15-2 Energy in Simple Harmonic Motion
30. An oscillating block–spring system has a mechanical energy of 1.00 J, an amplitude of 10.0cm, and a maximum speed of 1.20 m/s. Find (a) the spring constant,(b) the mass of the block, and (c) the frequency of oscillation.
31. A 5.00 kg object on a horizontal frictionless surface is attached to a spring with k 1000N/m. The object is displaced from equilibrium 50.0cm horizontally and given an initial velocity of 10.0 m/s back toward the equilibrium position. What are (a) the motion's frequency, (b) the initial potential energy of the block–spring system,(c) the initial kinetic energy, and (d) the motion's amplitude?
36. If the phase angle for a block–spring system in SHM is π/6 rad and the block's position is given by $x = x_m \cos(\omega t + \varphi)$, ,what is the ratio of the kinetic energy to the potential energy at time t = 0?

3rd Lecture:
Objective:
4. Describe how simple harmonic motion is the projection of uniform circular motion onto a diameter of the circle.

Chapter: 15
15-4: Circular motion (simple harmonic motion and uniform circular motion)

Application:
Module 15-4 Circular Motion
77. Figure 15-53 gives the position of a 20 g block oscillating in SHM on the end of a spring. The horizontal axis scale is set by t_s = 40.0 ms. What are (a) the maximum kinetic energy of the block and (b) the number of times per second that maximum is reached?
78. Figure 15-53 gives the position x(t) of a block oscillating in SHM on the end of a spring (t_s = 40.0 ms). What are (a) the speed and (b) the magnitude of the radial acceleration of a particle in the corresponding uniform circular motion?



4th Lecture:
1st Quiz

Objective:
1. From the wave function of a traveling wave, $y(x,t) = y_m \sin(kx - \omega t)$, prove that (i) $k = 2\pi/\lambda$, (ii) $\omega = 2\pi/T$ (iii) $v = +\omega/k$ and (iv) $v = -\omega/k$.

Chapter 16: Traveling and standing waves
16-1: Transverse waves (types of waves, transverse and longitudinal waves, wavelength and frequency, amplitude and phase, wavelength and angular wave number, period, angular frequency and frequency, phase constant, the speed of a traveling wave

5th Lecture:
Application:
Module 16-1 Transverse waves
1.If a wave y(x, t) = (6.0 mm) sin(kx + (600 rad/s)t + φ) travels along a string, how much time does any given point on the string take to move between displacements y = + 2.0 mm and y = - 2.0 mm?
5. A sinusoidal wave travels along a string. The time for a particular point to move from maximum displacement to zero is 0.170 s. What are (a) the period and (b) frequency? (c) The wavelength is 1.40 m; what is the wave speed?

7. A transverse sinusoidal wave is moving along a string in the positive direction of an x axis with a speed of 80 m/s. At t = 0,the string particle at x = 0 has a transverse displacement of 4.0 cm from its equilibrium position and is not moving. The maximum transverse speed of the string particle at x = 0 is 16 m/s.(a) What is the frequency of the wave? (b) What is the wavelength of the wave? If and $y(x,t) = y_m \sin(kx \pm \omega t + \varphi)$ is the form of the wave equation, what are (c) y_m, (d) k, (e) ω , (f) φ , and (g) the correct choice of sign in front of ω?

Objective:

2. Show that the speed of a transverse wave on a stretched string is $v = \sqrt{\frac{\tau}{\mu}}$, where τ is the tension on the string and μ is its mass per unit length.
[Or Prove that the speed of a wave along a stretched ideal string depends only on the tension and linear density of the string and not on the frequency of the wave.]

Chapter 16:
16-2: Wave speed on a stretched string (derivation from Newton's second law),

Application:
Module 16-2 Wave speed on a stretched string
14. The equation of a transverse wave on a string is $y = (2.0\ m) \sin[(20\ m^{-1})x - (600\ s^{-1})t]$. The tension in the string is 15 N. (a) What is the wave speed? (b) Find the linear density of this string in grams per meter.

6th Lecture:

Objective:
3. Wave functions of two sinusoidal transverse waves traveling along a stretched string are given by $y_1(x,t) = y_m \sin(kx - \omega t)$ and $y_2(x,t) = y_m \sin(kx - \omega t + \varphi)$, where φ is the phase difference between the interfering waves. Using the principle of superposition, find their resultant wave for different values of $\varphi = (0, \pi, 2\pi/3)$ and sketch a plot of the waves.
Chapter: 16

16-5: Interference of waves (the principle of superposition for waves, interference of waves),

Application:
Module 16-5 Interference of waves
32. What phase difference between two identical traveling waves, moving in the same direction along a stretched string, results in the combined wave having an amplitude 1.50 times that of the common amplitude of the two combining waves? Express your answer in (a) degrees, (b) radians, and (c) wavelengths.

7th Lecture:

Objective:
4. Two sinusoidal waves traveling in the opposite directions along a stretched string are given by $y_1(x,t) = y_m \sin(kx-\omega t)$ and $y_2(x,t) = y_m \sin(kx+\omega t)$. Using the principle of superposition, find the resultant standing wave. Find the positions of nodes and antinodes of the resulting standing wave. Sketch the figures when the waves are in phase and out of phase.

Application:
Module 16-7 Standing waves and resonance
53. A string oscillate according to the equation $y' = (0.50\ cm) \sin[(\pi/3\ cm^{-1})x] \cos[(40\pi\ s^{-1})t]$. What are (a) the amplitude and (b) the speed of the two waves (identical except for direction of travel) whose superposition gives this oscillation? (c) What is the distance between nodes? (d) What is the speed of a particle of the string at the position x = 1.5 cm when t = 9/8 s?

76. A standing wave results from the sum of two transverse traveling waves given by $y_1 = 0.050\cos(\pi x - 4\pi t)$ and $y_2 = 0.050\cos(\pi x + 4\pi t)$, where x, y_1,and y_2 are in meters and t is in seconds. (a) What is the smallest positive value of x that corresponds to a node? Beginning at t = 0, what is the value of the (b) first,(c) second, and (d) third time the particle at x = 0 has zero velocity?

8th Lecture:
2nd Quiz

Objective:
5. Derive expressions for resonant frequencies of a stretched string of length L with fixed ends (sketch figures).

Chapter 16:
16-7: Standing waves and resonance (standing waves, reflections at a boundary, standing waves and resonance)

Application:
Module 16-7 Standing waves and resonance
44. A 125 cm length of string has a mass 2.00 g and tension 7.00 N between fixed supports. (a) What is the wave speed for this string? (b) What is the lowest resonant frequency of this string?
49. A nylon guitar string has a linear density of 7.20 g/m and is under a tension of 150 N. The fixed supports are distance D = 90.0 cm apart. The string is oscillating in the standing wave pattern shown in Fig.16-39. Calculate the (a) speed, (b) wavelength, and (c) frequency of the traveling waves whose superposition gives this standing wave.
85. A 120 cm length of string is stretched between fixed supports. What are the (a) longest, (b) second longest, and (c) third longest wavelength for waves traveling on the string if standing waves are to be set up? (d) Sketch those standing waves.

9th Lecture:

Objective:
1.What are the wave front, plane wave and circular wave? Using a sketch, explain Huygens' principle.
2.What is a monochromatic light? Distinguish between coherent an incoherent light. What are the conditions of interference?
3. Derive the angular positions of maxima and minima for Young's double-slit interference experiment.
4. Derive the positions of mth maxima and minima for Young's double-slit interference experiment.

Chapter 35 and 36: Interference and diffraction

35-1: Light as a wave, Diffraction, 35-2: Young's interference experiment (diffraction, Young's interference experiment, locating the fringes)

Application:

Module 35-2 Young's Interference Experiment
14. In a double-slit arrangement the slits are separated by a distance equal to 100 times the wavelength of the light passing through the slits. (a) What is the angular separation in radians between the central maximum and an adjacent maximum? (b) What is the distance between these maxima on a screen 50.0cm from the slits?
20. Monochromatic green light, of wavelength 550 nm, illuminates two parallel narrow slits 7.70 mm apart. Calculate the angular deviation (θ in Fig. 35-10) of the third-order bright fringe (a) in radians and (b) in degrees.

10th Lecture:

Application:
Module 35-2 Young's Interference Experiment
21. In a double-slit experiment, the distance between slits is 5.0 mm and the slits are 1.0 m from the screen. Two interference patterns can be seen on the screen: one due to light of wavelength 480 nm, and the other due to light of wavelength 600 nm. What is the separation on the screen between the third-order bright fringes of the two interference patterns?
93. If the distance between the first and tenth minima of a double-slit pattern is 18.0 mm and the slits are separated by 0.150 mm with the screen 50.0 cm from the slits, what is the wavelength of the light used?
SP36-2: What is the distance on the screen between adjacent maxima (or minima) near the centre of the interference pattern? The wavelength λ of the light is 546 nm, the slit separation d is 0.12 mm, and the slit-screen separation D is 55 cm. Assume that θ is small enough to permit use of the approximation $\sin\theta \approx \tan\theta \approx \theta$, in which θ is expressed in radian measure.

Objective:
4. Prove that the intensity in double slit-interference is given by $I = 4I_0\cos^2\phi/2$ and the phase difference between two waves is given by $\varphi = (2\pi/\lambda)d\sin\theta$. Hence find the conditions for intensity maxima and minima. Plot a figure for I as a function of φ.

Chapter 35:
35-3: Interference and double-slit intensity (coherence, Intensity in double-slit interference)

Application:
Module 35-3 Interference and Double-Slit Intensity
29. Two waves of the same frequency have amplitudes 1.00 and 2.00. They interfere at a point where their phase difference is 60.0°.What is the resultant amplitude?

11th Lecture:
3rd Quiz:

Objective:
5. With appropriate figures, find the general equation of diffraction minima by a single-slit.

Chapter: 36
36-1: Single-slit diffraction

Application:
Module 36-1: Single-slit diffraction
2. What must be the ratio of the slit width to the wavelength for a single slit to have the first diffraction minimum at $\theta = 45.0°$?
7. Light of wavelength 633 nm is incident on a narrow slit. The angle between the first diffraction minimum on one side of the central maximum and the first minimum on the other side is 1.20°. What is the width of the slit?
9. A slit 1.00 mm wide is illuminated by light of wavelength 589 nm. We see a diffraction pattern on a screen 3.00 m away. What is the distance between the first two diffraction minima on the same side of the central diffraction maximum?